Mercy House

An evidence-grounded investigation world for language agents.

> The lamps have been turned up too high. Beneath the archive window, a woman lies motionless beside the desk.

1900. A mountain town. A detective with a personal history here.

10-minute presentation · Games & World Simulations / Open-compatible · Judges bring the detective.

The challenge

A plausible accusation is not a solved case.

Pressure	Required behavior
A threatening letter bears a name	Test authorship and provenance
A familiar subordinate handles a search	Preserve custody and examine omissions
A witness gives a frightened account	Separate observation, hearsay and inference
A personal connection suggests a story	Acquire support and revise conclusions

An incident that executes

Characters choose legal actions from their own situation. The engine commits movement, obstruction, assault and fabrication, then creates local perceptions and physical traces.

Recorded action tape → deterministic seeded reset → live investigation

We freeze the starting incident for reproducibility. We do not pretend a new LLM-generated crime is deterministic.

Current tape: four participants, three rounds. Other residents have authored routines; continuous autonomous lives are future work.

The detective controls the inquiry

Choose	Receive
Where to go	Local scene, routes and searchable features
Whom to question	Released recollections with source basis
Search yourself or delegate	Actual discoveries, omissions and custody
Which material to compare	Concrete correspondences, not a verdict
When to report	Supported coverage and false/unsupported penalties

No required-claim list, omniscient evidence catalog or instruction to suspect a named person.

Reward proof, not confident prose

```
R = clamp(C − 0.20 F − 0.10 U, 0, 1)
```

C: supported target coverage. F: distinct false findings. U: distinct unsupported findings.

A cloth match cannot identify a killer. Paper ownership cannot establish forgery. A correct accusation without acquired support scores zero.

Comparisons can compose into proof. Narrative uncertainty is retained separately; narrative itself is not automatically fact-checked.

Protect the answer from the solver

Role	Authority
Shell workspace	Notes, scripts, validated public town tools
Trusted harness	Budget, parsing, logging and orchestration
Protected world	Hidden state, provenance, transitions, reward
Model gateway	Credentials and configured model endpoints

Private character planner → gated recollections → public speaker. Raw hostile questions do not enter the private planner.

Actual local RL

Five skills: document comparison, witness choice, statement checking, following contacts, supervised delegation.

Policy	Successes	Rate
Untrained sampling	1 / 160	0.625%
Trained sampling	140 / 160	87.5%
Trained greedy	147 / 160	91.875%

20,000 CPU episodes · 8.8 minutes · 460 coefficients · no GPU · no training API calls.

Eighty held-out checkpoints, repeated twice. Same authored families. Public proposer supplies arguments; this is not learned language comprehension or LLM fine-tuning.

Live transfer is a separate result

2/5 completed live checkpoint exercises; small unpaired smoke, not whole-case success.

Exercise	Reward / 1
forgery	0.0
witness	0.0
statement_review	1.0
navigation	1.0
delegation	0.0

Frozen weights, real character/proposer calls. An interrupted partial attempt is retained and disclosed. Complete crime-scene detective runs so far have failed; checkpoint success is not a solved full case.

Evidence under stress

Check	Result
Five task reference/no-op pairs	1.0 / 0.0 for every task
Town + legacy tests	76 + 9 pass
Real-command isolation probe	Private routes denied; forged proof earns zero
Timeout recovery	Timed command followed by a valid action
Fresh no-cache local build	23.208 seconds

Sample success is explicitly scripted reference behavior. Failed model runs and raw logs are provided, not replaced with a staged score.

What this entry establishes

A reproducible world where supported reconstruction matters more than naming a culprit.

Runnable tools, private memories, grounded reward, observable failures and local RL learning.

Next: new causal plots and role assignments, repeated live transfer evaluations, controlled relationship interventions, and supernatural mechanics only when their causes are discoverable.

Limits: one canonical case, fixture training, limited report semantics, no whole-case live success yet. Final NPC deployment and organizer-machine timing must be configured and verified.